Trustworthy Machine Learning
Master's Programme in Data Science

Gender-Based Discrimination in Predicting Stroke Events

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Tiivistelmä — Referat — Abstract

Fairness in machine learning is essential, especially in sensitive applications like healthcare, where unfair predictions can amplify existing or historical biases. In this project, we study gender-based discrimination in the prediction of stroke events. Stroke prediction is a particularly sensitive task, since false negative errors may delay diagnosis and treatment, leading to serious consequences for patients. We trained two baseline models, a balanced logistic regression and an XGBoost classifier. Fairness was evaluated using Separation-based metrics, namely Equal Opportunity and Equalized Odds. Despite similar stroke prevalence for men and women, both models showed higher true positive rates for male patients, indicating systematic disadvantages for female patients. To mitigate these disparities, we applied three fairness-aware interventions: FN-aware thresholding, Equalized Odds constraints using the Fairlearn framework, and Equalized Odds threshold tuning. Our results show that strong in-processing constraints often lead to a collapse in predictive performance under severe class imbalance. In contrast, post-processing threshold tuning achieved the best fairness accuracy trade-off, substantially reducing gender gaps without compromising clinical usefulness.

Avainsanat — Nyckelord — Keywords

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1. Fairness Evaluation

Motivation

Our project focused on gender-based bias in stroke prediction. Since early detection is crucial for medical conditions as stroke, prediction errors directly impact patient care. Our goal was to see if a machine learning model, even when trained without gender as an input feature, would still show unfairness between men and women and whether this unfairness could be reduced using fairness-aware methods.

This topic felt particularly relevant because medical research has historically focused more on male patients, leading to potential gaps in how symptoms and risks are understood for women [4]. As a result, algorithms trained on such data may reflect or even amplify these existing biases.

Dataset and Preprocessing

We used the *Healthcare Stroke Prediction Dataset*, which contains 4,908 patient records. Each record includes demographic and clinical attributes and the target variable indicates whether a patient has experienced a stroke.

The dataset shows a mild gender imbalance (2,897 females and 2,011 males), but stroke prevalence is almost identical across genders: 4.14% for females and 4.43% for males. Therefore, any performance disparity between genders cannot be explained by true differences in prevalence.

Missing BMI values were removed, categorical variables were one-hot encoded, and continuous features were standardised. Importantly, gender was removed from the feature set during training. This choice allows us to study indirect discrimination, where bias arises through correlated variables rather than explicit use of the sensitive attribute. The data were split into training, validation and test sets using an 60/20/20 split.

1.1 Baseline Models and Fairness Metrics

We trained two baseline classifiers: a balanced logistic regression model with adjusted class weights to deal with label imbalance and an XGBoost classifier. To properly assess fairness and reliability, we turned to fairness metrics rooted in the notion of Separation, which compare error rates across demographic groups conditioned on the true outcome [6]. These metrics are especially suitable in our context because they assess whether, given that a person had a stroke, the model is equally likely to identify it correctly for both men and women. We focused on the following:

- **Equal Opportunity (EO)**: requires that the True Positive Rate (TPR) be equal across groups (here, male and female). This ensures that high-risk individuals are equally likely to be correctly identified, regardless of gender [3].
- **Equalized Odds (EOd)**: extends Equal Opportunity by also requiring that False Positive Rates (FPR) are equal. This enforce balance in both kinds of prediction errors [3].

We deliberately excluded Independence metrics, which require equal positive prediction rates regardless of actual stroke prevalence. This would be inappropriate here: men and women may have slightly different feature distributions or true prevalence, and in this context, forcing parity without regard for risk can lead to over or under treatment [2].

We now present the baseline results:

Model	**Accuracy**	**TPR Gap**	**FPR Gap**
Logistic Regression (balanced)	0.743	+0.073	+0.01
XGBoost	0.715	+0.113	+0.02

Table 1.1: Baseline results showing gender disparities in error rates.

Note that all results presented are the bootstrapped means to increase robustness and mitigate influence of a specific train test split. We also present plots with visualized confidence intervals in the appendix.

Both models achieved solid predictive performance, considering the unbalanced dataset. However, the observed gaps in TPR show that male patients were more likely to be correctly identified as stroke cases compared to female patients or, consequently, that women were more often subject to False negatives cases. This is a real issue, as it can lead to delayed treatment, lack of further screening, or missed opportunities for prevention.

While False positives (predicting stroke when there is none) are less dangerous clinically, they still carry consequences, such as psychological stress, unnecessary follow-ups, and increased healthcare costs. For this reason, fairness metrics that consider both TPR and FPR (like Equalized Odds) are relevant, but our main focus remains on reducing excess false negatives for women.

These results demonstrate that despite removing gender from the input features, both models still learned patterns that led to indirect discrimination. This motivated us to explore fairness-aware interventions in the second part of the project.

2. Fairness Interventions

In the context of stroke prediction, false negatives are particularly dangerous. They represent missed opportunities to identify high-risk patients, potentially delaying essential diagnostic procedures or preventive care. Because stroke outcomes depend heavily on early intervention, any systematic bias that increases the false negative rate for a specific group, such as women, can reinforce existing inequalities in healthcare access and treatment [4].

Given this, our fairness goal was to reduce disparities in sensitivity (TPR) and false negative rates between men and women, while still maintaining reasonable model performance.

Although fairness aware pre-processing techniques such as reweighing and sampling based data balancing are commonly used to mitigate discrimination before model training [5], we chose not to apply them in this project. The low overall prevalence of stroke cases (around 4%) made reweighting unstable in our data and, since gender is correlated with other features, indirect discrimination can persist even after reweighting. So pre-processing alone is unlikely to fully address this type of bias without distorting the underlying data distribution.

Instead, we focused on model-level interventions: Post-Processing methods that modify predictions after training, and In-Processing methods that introduce fairness constraints during training.

2.1 Method 1: FN-Aware Thresholding

FN-aware thresholding is post-processing method that aims to reduce the False Negative Rate (FNR) for disadvantaged groups without retraining the model. The idea is to apply a different classification threshold for each group: we kept the male threshold fixed at 0.5 and searched for a lower threshold for females to increase their sensitivity (TPR) [3], this can be done on the validation set .

This is based on the assumption that false negatives (predicting no stroke when one is actually present) are more harmful than false positives in our specific settings. Missing a stroke case can delay diagnosis and treatment, while a false alarm may only

lead to extra testing. Therefore, prioritising sensitivity for women, who were under identified in baseline models, is a meaningful fairness goal in this context.

Model	Accuracy Drop	TPR (M)	TPR (F)	Gap
Logistic Regression	-0.051	0.844	0.868	-0.024
XGBoost	-0.033	0.807	0.796	0.011

Table 2.1: Results of FN-aware thresholding.

The results show that this method was able to significantly reduce the TPR gap (from 0.073 to -0.024 for LR, so actually slightly higher TPR for women, and from 0.113 to 0.011 for XGBoost). This came at the cost of reduced overall accuracy, however, the accuracy drop was relatively small and may be acceptable depending on clinical priorities.

In the end, this was the best result for both models, that still maintained reasonable accuracy and lead to elimination of the TPR gap. Note that the false positive rates are now higher for women as well, which, as we discussed, also has consequences, and this trade-off should be kept in mind.

2.2 Method 2: Equalized Odds

Equalized Odds is a fairness criterion that requires both the True Positive Rate (TPR) and the False Positive Rate (FPR) to be equal across sensitive groups [3]. In our context, this means that the model should perform similarly for male and female patients in terms of correctly predicting stroke cases (TPR) and avoiding false alarms (FPR).

We implemented this constraint using the Exponentiated Gradient algorithm from the `fairlearn` python package [1]. This method trains a model by combining several classifiers into a weighted ensemble. During training, it solves a constrained optimisation problem that balances two goals: minimising prediction error and reducing fairness violations. Instead of modifying the data or applying post-processing, it directly adjusts the learning process to satisfy fairness criteria.

Model	Accuracy Drop	TPR (M)	TPR (F)	Gap
Logistic Regression	-0.402	0.909	0.857	0.052
XGBoost	+0.102	0.218	0.133	0.085

Table 2.2: Results under Equalized Odds constraints.

The results show that Equalized Odds also slightly reduced the TPR gap. However, the price were collapsing models: for LR, the model produced almost all positive

predictions. For XGBoost, the drop in sensitivity was less extreme, but this model suffered the opposite problem of predicting almost all zeros. This shows a trade-off between fairness and usefulness. Although Equalized Odds improves error parity, it may not be practical when the dataset is heavily imbalanced and high sensitivity is needed. In clinical contexts, missing positive cases is dangerous, so models with low TPR are not acceptable. For this reason, we explored a third method focused on adjusting thresholds after training to improve fairness in a different way.

2.3 Method 3: Equalized Odds Threshold Tuning

This method builds upon the ideas of Equalized Odds but offers a more flexible and practical solution in cases where full EO constraints are too strict. We investigated the probabilities returned by the models in Method 2, and deemed that the default threshold of 0.5 was not appropriate due to the models returning more extreme probabilities. The goal is to find a threshold common for both genders that equalises TPRs between men and women, while keeping reasonable performance. Since it operates post hoc and does not require retraining the model, it also preserves the original model structure and decision boundaries.

Model	**Acc. Drop**	**TPR (M)**	**TPR (F)**	**Gap**
Logistic Regression	-0.330	0.835	0.802	0.032
XGBoost	-0.004	0.786	0.742	0.043

Table 2.3: Equalized Odds threshold tuning results.

Compared to the previous method (full Equalized Odds using Fairlearn), this approach achieved somewhat better overall performance for logistic regression, however, the accuracy remained small. For XGBoost, the TPR gap was considerably reduced, almost completely closing the TPR gap with virtually no loss in accuracy. However, when we look at the confidence intervals obtained using bootstrapping (plotted in the appendix), we see that they are very wide for the XGBoost TPR gap, and so the logistic regression with threshold tuning remains our strongest method for TPR gap reduction for both models.

3. Conclusion

This project aim to explore whether gender bias could emerge in stroke prediction models, even when gender was not explicitly used during training. Our analysis confirmed that indirect discrimination was present: Female patients were more likely to receive false negative predictions, a particularly serious issue in domain where early detection is essential.

To address this, we applied three fairness-aware methods and compared their performance. FN-aware thresholding offered an intuitive post-processing fix, improving sensitivity for women but slightly lowering overall accuracy. In the end, this approach offered the best trade-off between fairness and utility. Equalized Odds constraints successfully reduced disparities but often led to models that were too conservative, with poor accuracy or sensitivity due to class imbalance. Another promising solution was Equalized Odds threshold tuning. It achieved almost equal true positive rates between genders for XGBoost, without sacrificing accuracy or model interpretability, but the results were not very robust. On the other hand, the accuracy of logistic regression for this method remained poor.

Our findings highlight the importance of fairness evaluation in healthcare applications of machine learning, In the case of stroke prediction, we saw how these disparities mainly affected women through higher false negative rates. While no method gave a perfect solution, some of the simpler post-processing techniques helped reduce the gap without hurting performance too much. This suggests that fairness-aware adjustments can be helpful, especially when chosen carefully for the context.

Use of AI Tools

We used AI tools (ChatGPT) to fix minor bugs in the code and to revise the writing in the final report. Ideas, experiments, and analysis were our own.

Visual Summary of Results

We present a selection of plots to visually summarise key findings from our analysis. These include demographic distributions, baseline disparities, and the trade-offs between fairness and accuracy for different fairness interventions. We also include threshold curves for the baseline models to visualize how the TPR and FPR for woman changes when we modify the threshold to achieve similar TPR as for male patients.

Dataset Overview

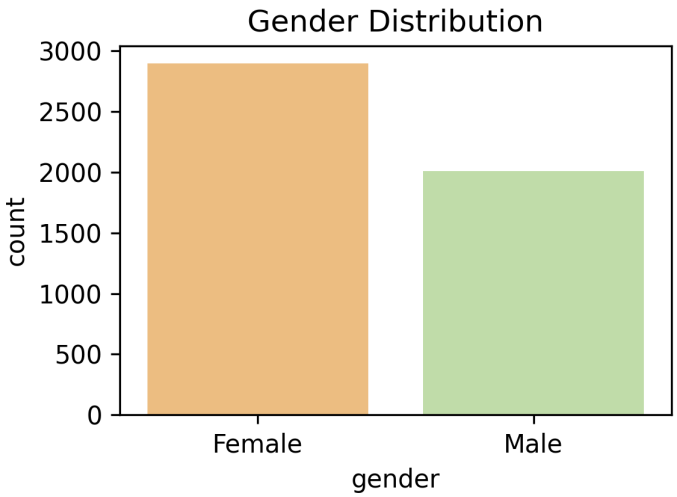

Figure 3.1: Gender distribution in the dataset.

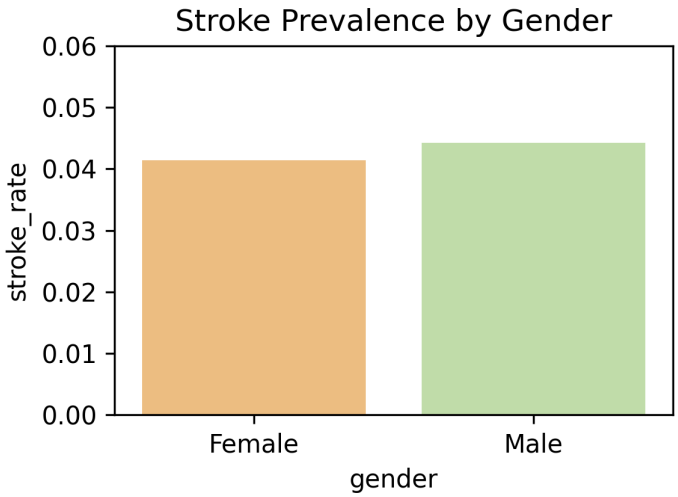

Figure 3.2: Stroke prevalence is nearly equal for both genders.

Logistic Regression Results

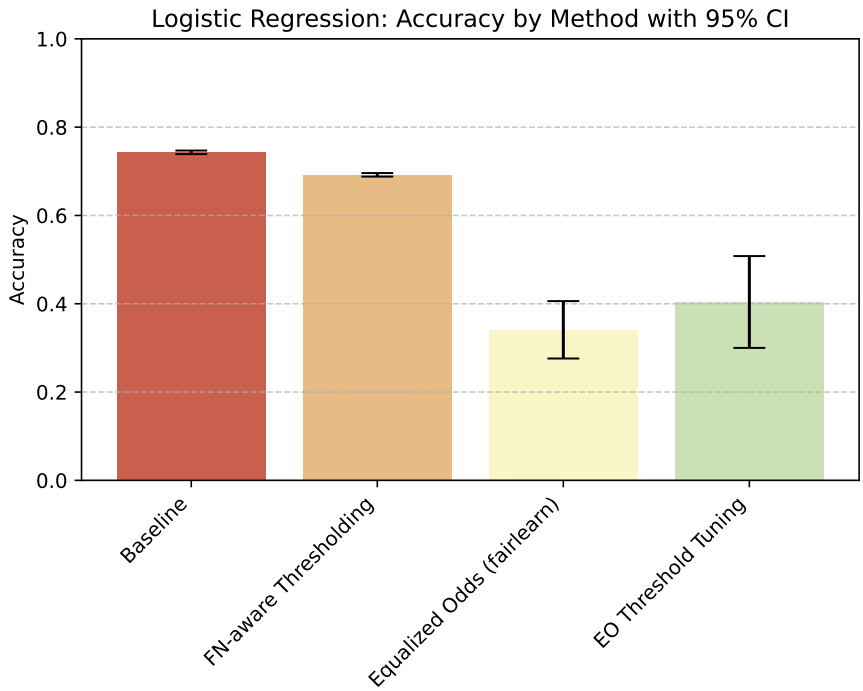

Figure 3.3: Accuracy of logistic regression under different fairness methods.

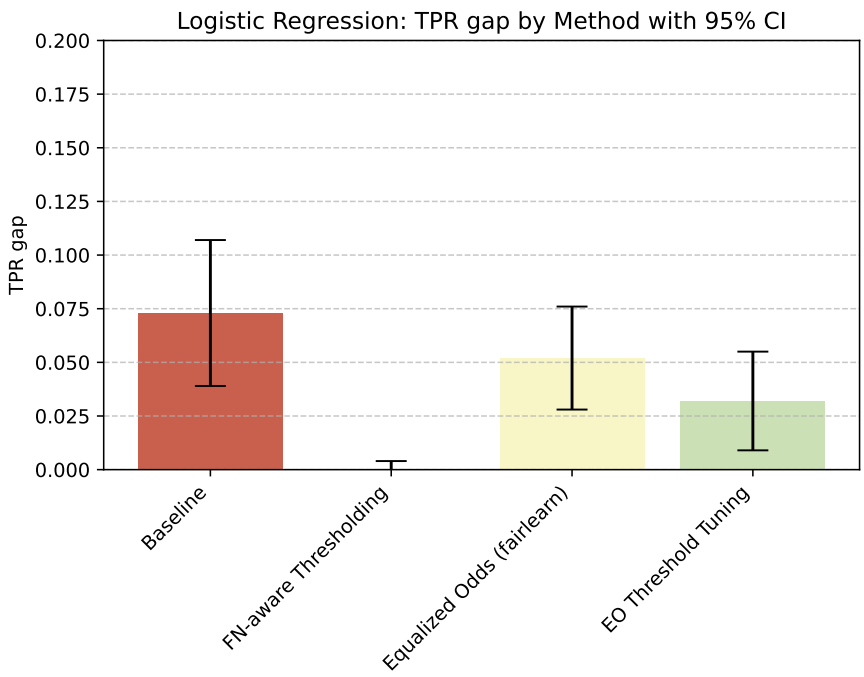

Figure 3.4: TPR gap for logistic regression under different methods.

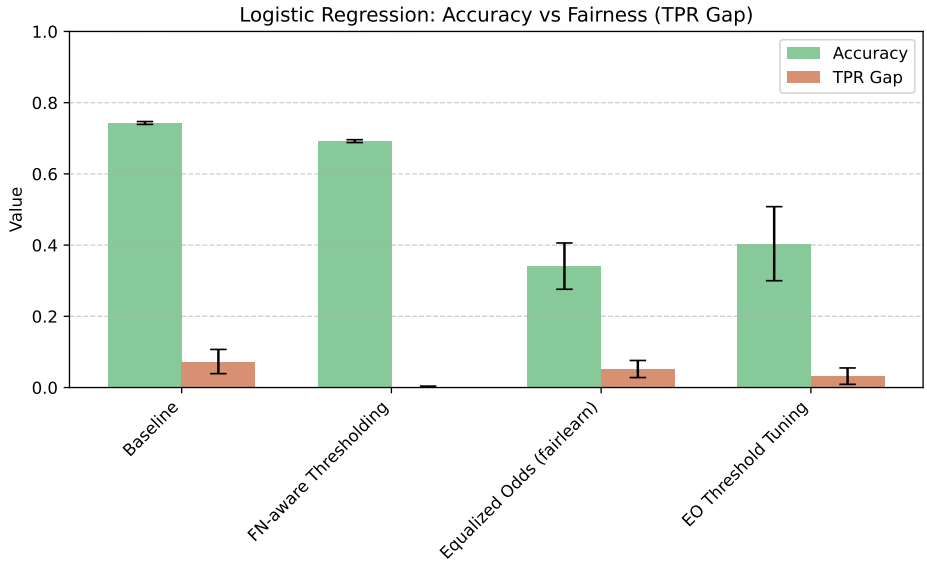

Figure 3.5: Fairness-accuracy trade-off for logistic regression.

XGBoost Results

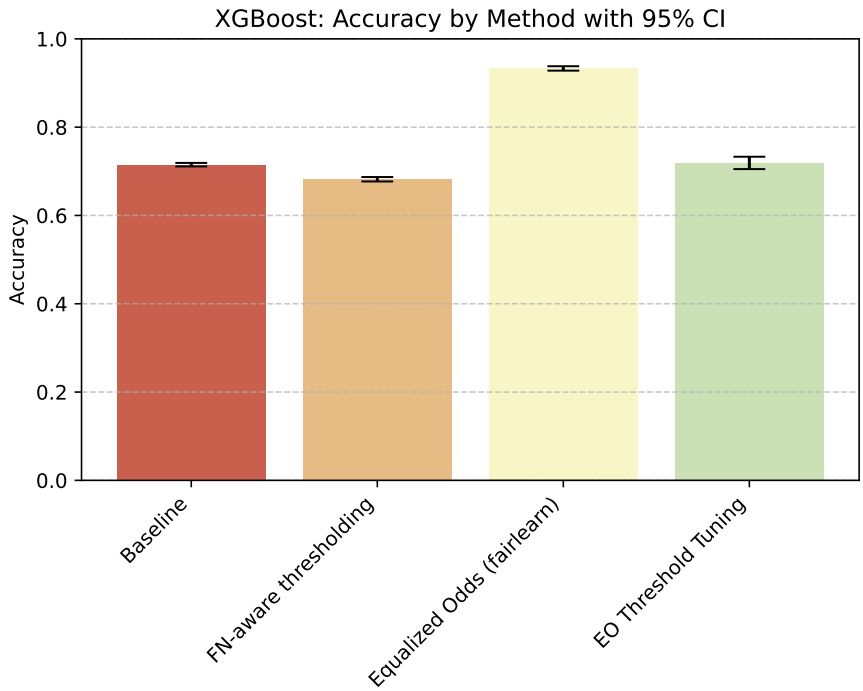

Figure 3.6: Accuracy of XGBoost under different fairness methods.

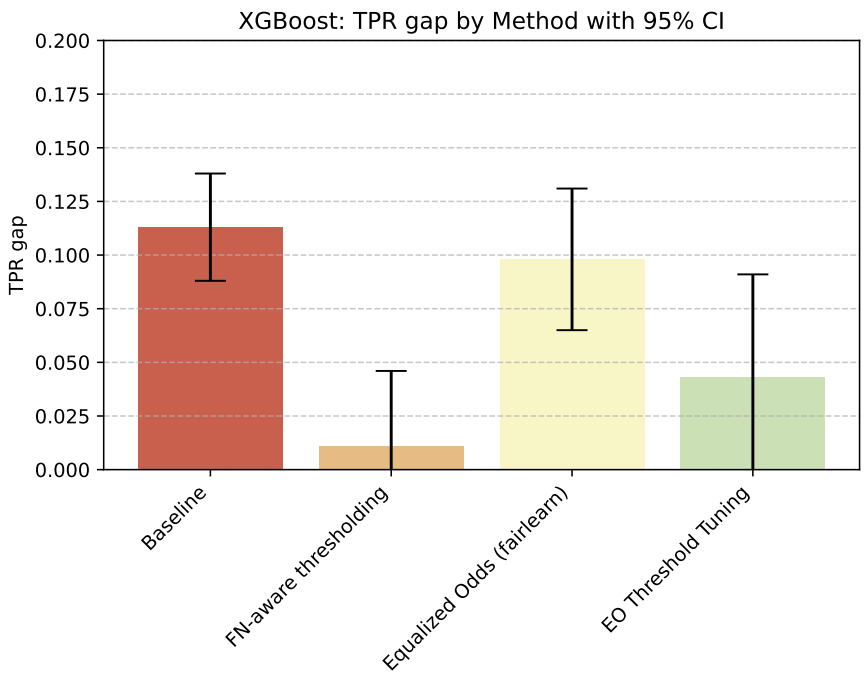

Figure 3.7: TPR gap for XGBoost under different methods.

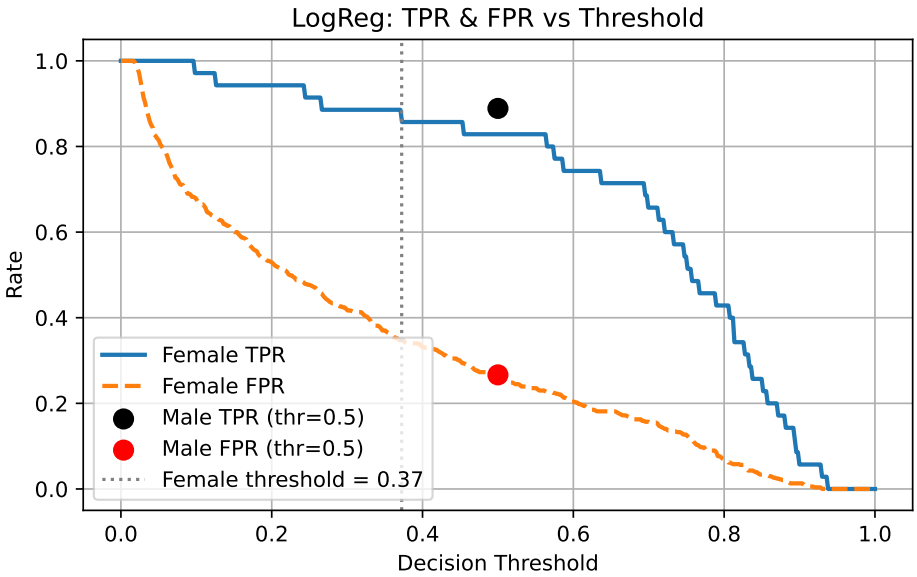

Figure 3.8: Threshold curves for Logistic Regression

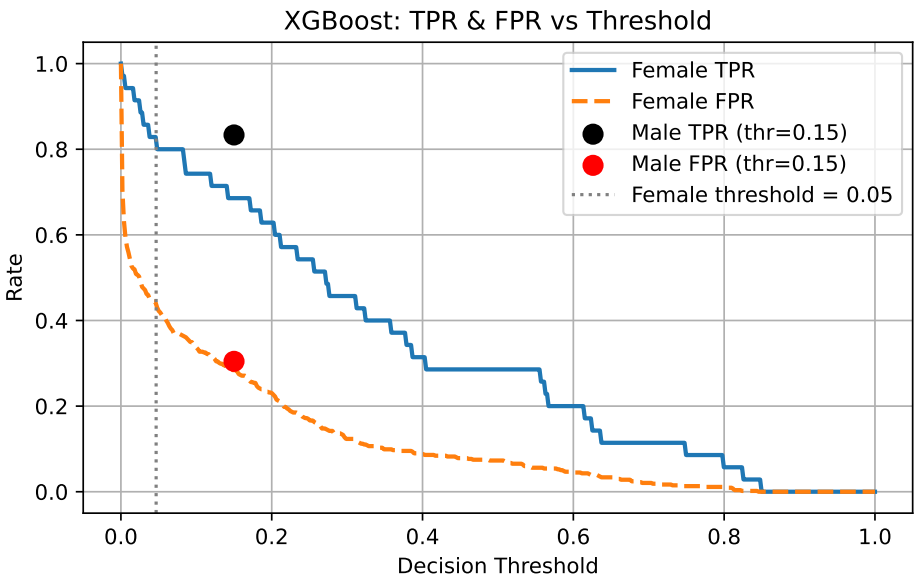

Figure 3.9: Threshold curves for XGBoost

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